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Auditing Model Substitution in LLM APIs: Why Software Tests Fail and Hardware Attestation Wins

Commercial LLM API providers may covertly substitute cheaper models;
software-based detection is unreliable — Trusted Execution Environments offer a provably secure alternative.

For ML Researchers · AI Security Practitioners · Cloud Infrastructure Engineers

Source: arxiv.org/abs/2504.04715

The Model Substitution Problem: Economic Incentives Drive Covert Swaps

Providers face strong incentives to serve cheaper models while charging for premium ones

The Core Problem

APIs are black boxes — users cannot verify which model actually processes their queries.

Providers charging for Llama-3-70B can silently serve a quantized variant or smaller model.

GPU cost savings can be substantial (3–8× reduction), creating strong economic pressure.

Formally modelled as hypothesis testing:

H_0 : API serves $P_{\text{spec}}(y|x)$ — the advertised model

H_1 : API serves $P_{\text{actual}}(y|x)$ — a substitute model

Four Substitution Types

① Smaller Model

8B served instead of 70B

② Quantized Variant

FP8/INT8 instead of FP16 weights

③ Fine-tuned Version

Updated or distilled checkpoint

④ Different Family

Entirely different architecture

FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

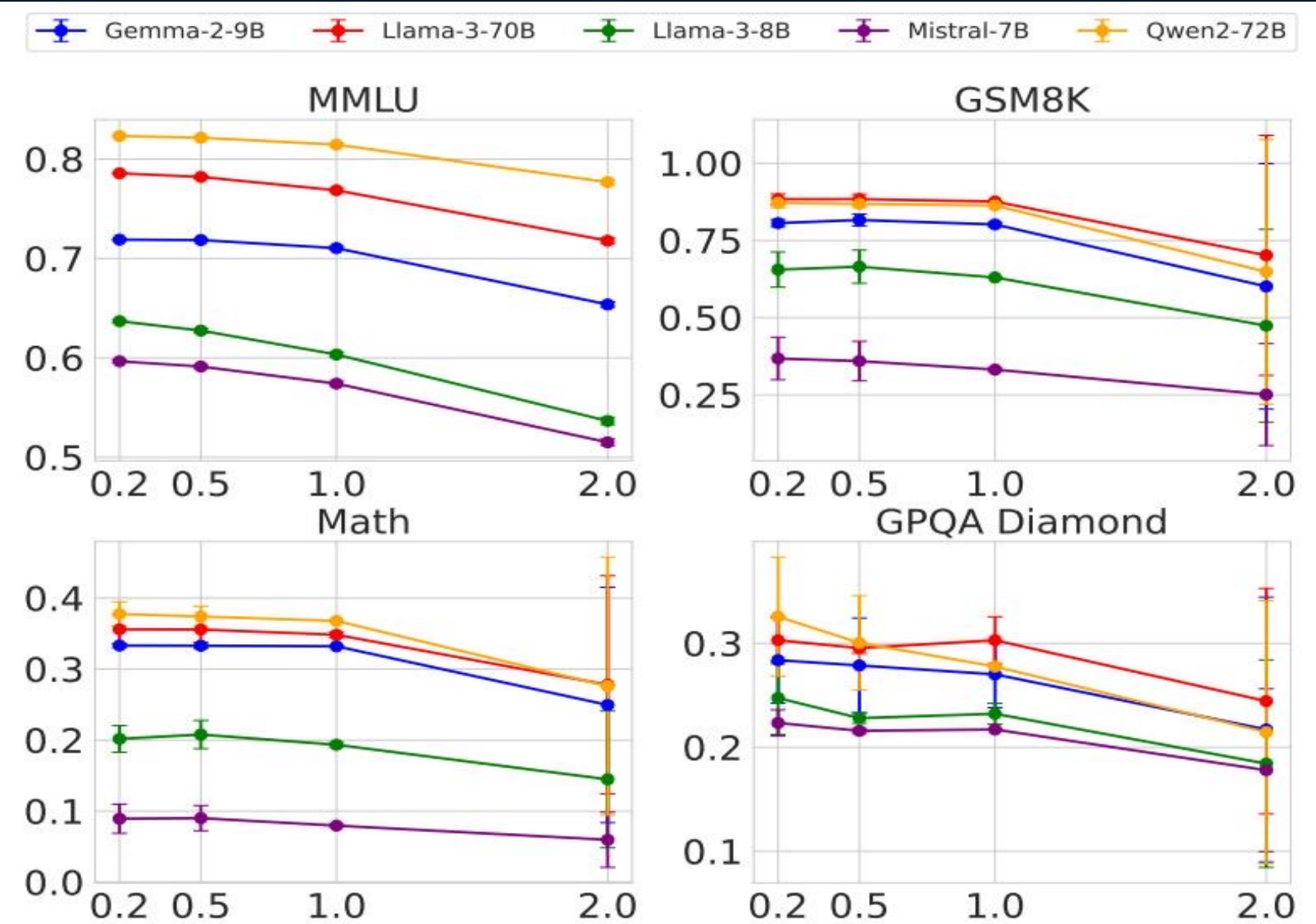
Benchmark scores for original vs FP8-quantized variants across five model families

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct-FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct-FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it-FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct-FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3-FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

★ For most models, FP8 scores fall within error bars of the originals (especially MMLU & GSM8K), making covert substitution statistically undetectable with practical query budgets.

Higher Temperature Amplifies Performance Gaps but Increases Noise

Temperature scaling complicates detection — degrading all models while narrowing observable gaps



Key Observations

Temperature 0.2

Near-deterministic output; original vs FP8 gap is small but measurable.

Temperature 1.0

Standard sampling; distributions overlap substantially.

Temperature 2.0

Performance collapses for all models — the substitution signal drowns in noise.

Detection limit

Higher T lowers SNR, requiring exponentially more queries to achieve statistical significance.

Source: Figure 1 — arxiv.org/abs/2504.04715

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

Same model, different infra → indistinguishable variation from actual substitution

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Why Fingerprinting Breaks

Software Drift

transformers 4.40 vs 4.50 and vLLM 0.6.2 vs 0.8.2 produce measurably different log probabilities for identical inputs.

Hardware Variation

A100 vs H100 GPU arithmetic introduces floating-point non-determinism across batches.

False Positives

An auditor cannot distinguish legitimate infrastructure variance from an actual model substitution.

Consequence

Log-prob fingerprinting is fundamentally unreliable as a production verification tool.

Source: Figure 5 — arxiv.org/abs/2504.04715

Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive

Mixing substitute responses at probability p drives the observable distribution toward the legitimate model

The Randomized Substitution Attack

Mechanism

Provider routes each query to the genuine model with probability $(1 - p)$ and to the substitute with probability p .

Mathematical consequence

The observed mixture distribution converges to P_{spec} as $p \rightarrow 0$. Smaller $p \implies$ exponentially larger query budget required to reject H_0 .

Benchmark evasion

Provider detects known benchmark prompt patterns and routes them exclusively to the genuine model, defeating accuracy-based audits entirely.

Query budget explosion

At $p = 0.05$, a statistically valid detection test may require millions of API calls — economically and practically infeasible for any auditor.

Statistical Detection Complexity

$p = 0.50$	50% substitute	Detectable
<i>~hundreds of queries</i>		
$p = 0.20$	20% substitute	Difficult
<i>~thousands of queries</i>		
$p = 0.10$	10% substitute	Very Difficult
<i>~tens of thousands</i>		
$p = 0.05$	5% substitute	Practically Infeasible
<i>~millions of queries</i>		

As $p \rightarrow 0$, $P_{\text{mixed}} \rightarrow P_{\text{spec}}$ — detection becomes statistically impossible.

Software-Only Auditing Methods Are Fundamentally Unreliable

Comparison of three detection approaches and their critical failure modes

Method Type	Core Approach	Weakness	Practical Limitation
Text-Based Statistical Tests (e.g., MMD)	Compare output text distributions between reference and API using maximum mean discrepancy or similar tests.	Benchmark scores overlap within error bars for FP8 quantization. High noise at practical temperatures.	Requires impractically large query budgets. Fails entirely against subtle or randomized substitutions.
Log-Probability Fingerprinting	Compare token-level log-probabilities between a local reference model and the API response.	Production nondeterminism: different software versions (transformers, vLLM) and GPU hardware (A100 vs H100) produce divergent log-probs.	Cannot distinguish legitimate infra variation from actual substitution. High false-positive rate renders it unusable.
Adversarial Countermeasures (Evasion)	Provider detects benchmark prompts or randomizes routing to minimize detectable signal.	Benchmark evasion routes known test prompts to genuine model; randomized routing dilutes substitution signal.	No software audit can overcome an actively adversarial provider. Query budgets become astronomical ($p \rightarrow 0$).

Conclusion: No software-only method can provide reliable guarantees against a motivated, adversarial provider.

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

Hardware attestation closes the verification gap that software methods cannot bridge

What Are TEEs?

Trusted Execution Environments are hardware-secured enclaves that provide cryptographic attestation of both the code being executed and the data being processed.

NVIDIA Confidential Computing on H100 GPUs enables TEE-protected inference today, with only modest performance overhead compared to standard serving.

TEE Guarantees

Provable	Cryptographic proof that correct model weights are loaded — not just a claim.
Efficient	Modest overhead vs ZKPs which are computationally prohibitive at LLM scale.
Robust	Cannot be defeated by adversarial routing or benchmark evasion strategies.

Comparison: Software Methods vs TEEs

Property	Software Methods	TEEs
Provability	✗ No	✓ Cryptographic
Adversary-proof	✗ No	✓ Yes
Query budget	✗ Millions+	✓ Zero queries
Nondeterminism	✗ Confounded	✓ Not applicable
Availability	✓ Today	✓ H100 today
Overhead	✓ Minimal	⚠ Modest

Key Takeaways: Close the Verification Gap with Hardware Attestation

Five findings and a call to action for the ML security community

01

Model Substitution Is Economically Motivated

GPU cost savings of 3–8× create strong provider incentives to serve cheaper models. The API black-box makes this trivially exploitable without external verification.

02

FP8 Quantization Is Nearly Undetectable via Benchmarks

Performance drops are typically within error bars on MMLU and GSM8K. Benchmark-based auditing cannot reliably flag quantization substitution.

03

Inference Nondeterminism Defeats Log-Probability Fingerprinting

Software version changes and GPU hardware differences produce variation indistinguishable from substitution — rendering metadata-based detection unreliable.

04

Randomized Routing Defeats Statistical Tests

At low substitution rates ($p \rightarrow 0$), the required query budget grows exponentially. Benchmark evasion eliminates even this impractical avenue.

05

TEEs Are the Only Viable Path to Verifiable LLM APIs

Hardware attestation via NVIDIA H100 confidential computing provides cryptographic proof of correct model execution — practical, deployed, and adversary-resistant today.

Call to Action: Adopt TEE-based attestation as the standard for LLM API procurement contracts, cloud SLAs, and compliance frameworks. Demand cryptographic proof, not promises.